SURYA ARAKALGUD MAHESH

Embedded Software Engineer | Networked Devices | Automotive & Control Systems

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PROFESSIONAL SUMMARY

Embedded software engineer with 2+ years of production experience on automotive ECU/SoC platforms and a current volunteer engineering role on a networked patient-repositioning platform. Hands-on with C/C++, Python, ARM-based systems, Linux, CAN and Ethernet/TCP-IP, ECU bring-up, OTA/diagnostics, fault management and recovery validation. Experienced in hardware-software debugging, packet analysis, automated regression and embedded control fundamentals.

CORE COMPETENCIES

Languages & Tools: C/C++, Python, Bash, CMake, Makefile, cross-compilation, Git, Jenkins, Docker, Pytest
Automotive Embedded: ECU/SoC platforms, ARM Cortex-A53/Cortex-M7, AUTOSAR, CAN, Ethernet/TCP-IP, OTA/diagnostics, fault management, fault injection, recovery validation
Real-Time / RTOS: interrupts, DMA, mutexes/semaphores/queues, priority inversion, race conditions, watchdogs, deadlines, jitter, deterministic execution
BSP / Drivers / Peripherals: BSP/HAL concepts, driver integration, GPIO, timers, PWM, ADC, UART, SPI, I2C, CAN, pin-mux, memory-mapped I/O, U-Boot/bootloader
Motor-Control Fundamentals: EV powertrain, inverter/PWM, ADC-PWM synchronization, FOC architecture, PI loops, regenerative braking, derating, sensorless/back-EMF concepts
Debugging & Validation: ECU/test-bench validation, JTAG/SWD concepts, oscilloscope/logic analyzer, UART/serial, Wireshark, tcpdump, gdb, perf, ftrace, regression testing
Networking & Systems: Socket APIs, Linux kernel/driver integration, NETCONF, RESTCONF, PyATS/Genie, Netmiko, iPerf, Spirent, Cisco IOS-XE/NX-OS

PROFESSIONAL EXPERIENCE

Q's Ministry / BHITS — *Oct 2026 - Present*

Embedded Systems, Networked Device & Control Engineer (Volunteer – Los Angeles)

- Assigned to embedded firmware and control development for a networked four-winch patient-repositioning platform, including C/C++, coordinated control state machines, PWM/timing, GPIO, watchdogs and fault recovery.
- Initial focus on reviewing existing firmware, schematics, control logic and test records to document the technical baseline; role scope includes device communications, APIs, telemetry and multi-station management.
- Responsibilities include hardware-team collaboration, controller interface specifications, board bring-up support, test automation and fault-oriented validation of communication loss, restarts and safe-state recovery.
- Assigned networked-device scope includes unique station identification, registration/discovery, authenticated LAN/Wi-Fi communications, command acknowledgments, status reporting and reconnect behavior.
- Control-system responsibilities include manual/app control arbitration, current and limit sensing, startup/shutdown behavior and local safety limits, with fallback behavior for network or application failures.

Bosch Global Software Technologies — *Sep 2022 - Aug 2024*

Software Engineer, Embedded Networking & Automotive Systems

- Supported bring-up and communication validation across a six-SoC/ECU automotive test bench using ARM Cortex-A53/Cortex-M7 in AUTOSAR environments; verified cross-node signals and isolated boot/configuration issues.
- Developed C/C++ and Python software for Linux-based ECU OTA and diagnostic pipelines; implemented fault-triage logic and execution tracing that reduced manual debugging effort by 60%.
- Owned an ECU fault-management module, implementing fault detection/handling and validating recovery behavior through controlled fault injection.
- Built ECU fault-injection, regression and latency/stability KPI workflows, improving incident resolution time by 30% and system stability by 20%.
- Diagnosed CAN and Ethernet/TCP-IP failures using traces, packet analysis and configuration checks; resolved initialization, synchronization and TCP/IP issues, improving communication reliability by 40%.
- Built C/Python embedded network middleware using raw sockets and CMake; profiled CPU, memory and latency with perf/cProfile, reducing average packet-processing latency by 30%.
- Developed a Linux virtual network device kernel module with Tx/Rx paths, interrupt handling and DMA-style buffering, plus a CAN/TCP-IP fault-injection framework to benchmark recovery, packet loss and stability.

EDUCATION

California State University, Long Beach — M.S. Information Systems — *Aug 2024 - May 2026*

Relevant coursework: AI Systems, Applied Analytics, Software Engineering, Data Systems

Visvesvaraya Technological University— B.Tech, Electronics & Communications Engineering — *Aug 2018 - Aug 2022*

Relevant coursework: Embedded Systems, Digital Signal Processing (DSP), Network Theory, Signals and Systems